

Experiment I: AC Induction & Inductance

PART 1: AC induction and magnetic coupling

1.1 Objective

1. To observe principle of AC induction
2. To observe how energy is transferred by electromagnetic induction.

1.2 Brief Theory

When the flux inside a coil varies, voltage E_1 is induced across its terminals (Faraday's Law). If the flux varies sinusoidally, the induced voltage is AC with RMS value given by

$$E_{1rms} = 4.44 f N_1 \phi_{max} \quad (1)$$

where, E_{1rms} is the induced voltage in Volts, f is the frequency of the alternating flux in Hz (50 Hz for our case), N_1 is the number of turns in the coil, and ϕ_{max} is the peak value of the flux in Wb. If the induced voltage and the number of turns are known, peak flux can be calculated from

$$\phi_{max} = \frac{E_{1rms}}{4.44 f N_1} \quad (2)$$

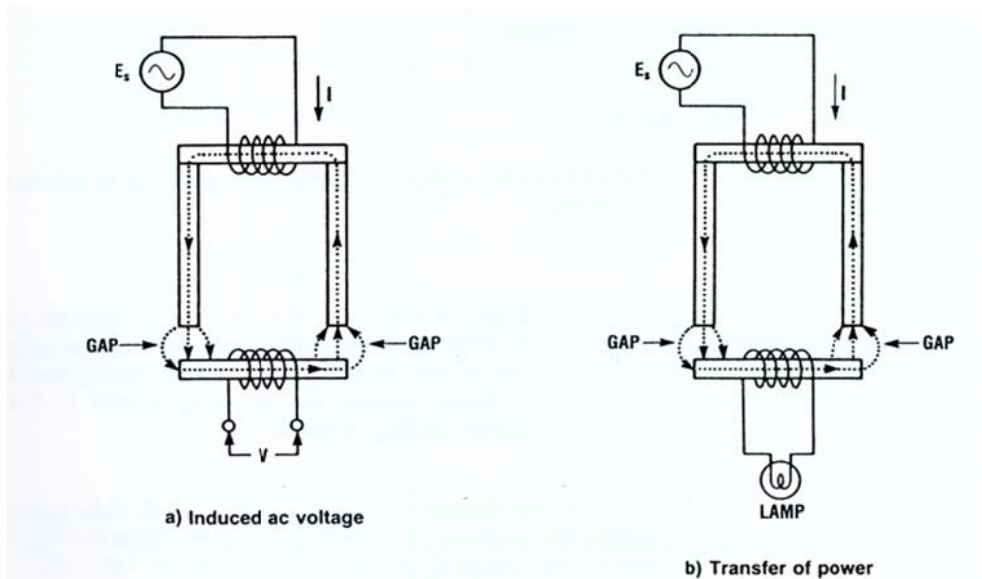


Fig. 1 Energy transferred through air gaps

If a second coil is present, as in the lower part of the magnetic circuit shown in Fig. 1, an AC voltage will be induced across its terminals. Assuming that there is no leakage of flux along the core, i.e., the same flux links both coils, the RMS value of the voltage induced in the second coil is given by (3).

$$E_{2rms} = 4.44 f N_2 \phi_{max} \quad (3)$$

When a load resistor is connected across the second coil, the induced voltage causes a current to flow in this resistor. Thus, it is possible to transfer energy between the two coils by electromagnetic induction.

Equipment required

2 x AC voltmeters	AC ammeter	Lamp
AC power supply (Variac)	1x Oscilloscope	Current probe
Connection leads	2 x Coils (A and B) with multi turns	Mounting base
3 x 133mm Laminated bars (indicated as ① in figures).		
133 mm laminated bar with hook (indicated as ③ in figures),		
178 mm laminated bar (indicated as ④ in figures).		
Non-magnetic 0.75 mm thick Fibrous spacers (indicated as ⑧ in figures)		

CAUTION

Do not make any connection or change to circuit when power supply is on.
You might need to select appropriate scales for ammeter and voltmeter.
While changing from one scale to another in any meter, turn off the power supply first. Do not disconnect/connect anything while power is on.

1.3 Procedure: Electromagnetic induction and magnetic coupling

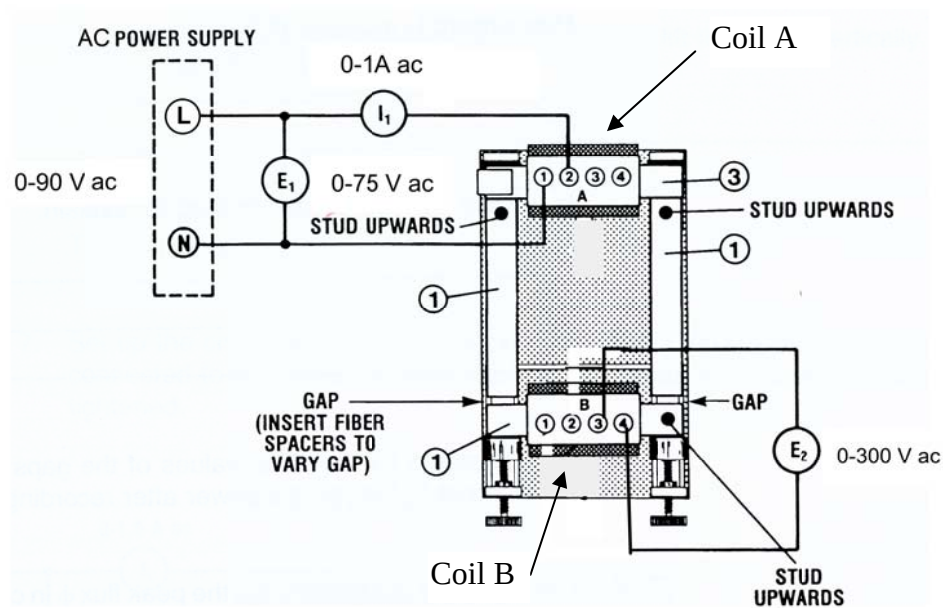


Fig. 2 Circuit to observe the effect of air gaps on AC induction and magnetic coupling

1. Set up the circuit shown in Fig 2. Tighten the screw to reduce air-gaps to a minimum.
2. On the AC supply ensure that control knob is set at zero. Turn on the power supply and **slowly** raise voltage E_1 to 50 V(rms) ac using the Variac. Measure voltage E_1 , current I_1 and voltage E_2 and record them in your logbook/notebook.

Turn off the power.

- Turn the two screws on the mounting base counter-clockwise to slacken the bar inside the coil B. Insert a 0.75 mm Fibrous spacer to create two gaps of length 0.75 mm. The fibre spacer passes inside the coil. Tighten the screws.

Turn on the power and **slowly** increase the voltage E_1 to 50V ac. Measure I_1 and E_2 and record result in **Table 1**. Use **0-300 V scale in E_2 voltmeter**.

Turn off power after bringing the voltage down to zero in the Variac.

- Repeat step 3 for the other values of the gaps shown in Table 1 and tabulate these as indicated in Table 1 below. Do not forget to turn off power before you change any spacer.

Table 1

Gap (mm)	E_1 (V)	I_1 (A)	E_2 (V)	Peak flux ϕ (from Eq 2.3)
0	50			
0.75	50			
1.5	50			
3	50			

Question 1. Explain how I_1 changes with the size of gap lengths and why. (write your answer in your logbook.

1.4 Energy transfer

- Set up the circuit of Fig 3 with no gaps and a lamp connected to terminals 3 and 4 of coil B. Make sure that the screws are tightened so that no air gap between the laminated bars.
- Turn on the power supply, and gradually raise the voltage E_1 to 50 V ac. Measure I_1 , E_2 and I_2 and record the results in the Table 2.. Since current I_2 is too small to measure with the ammeter, use the current probe and oscilloscope. Note down the rms value from the oscilloscope.

Turn off the power.

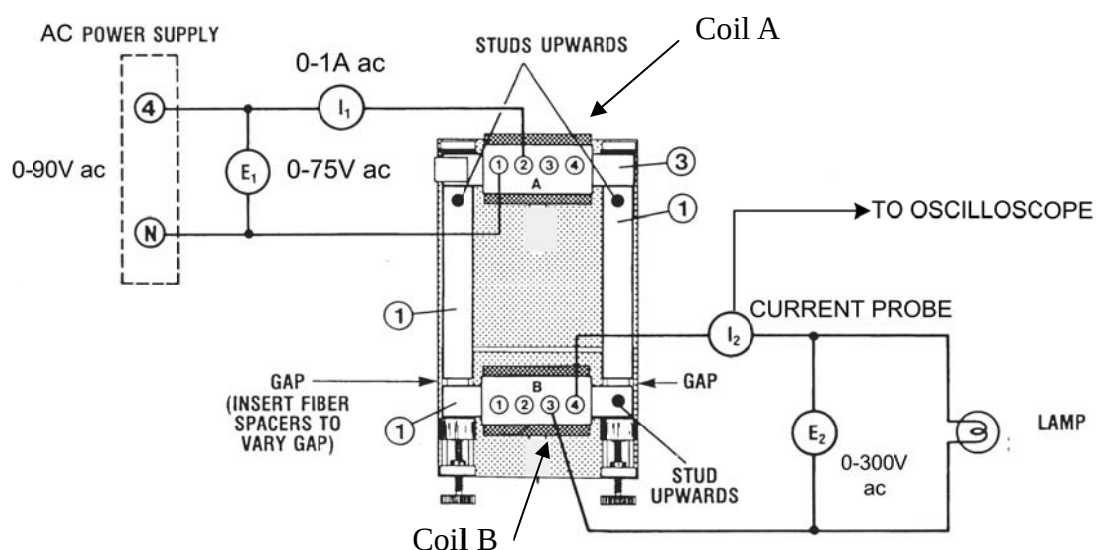


Fig 3 circuit to observe the energy transferred through air gaps.

CAUTION: The lamp could be very hot, avoid touching.

7. Loosen the bar inside the coil B and insert the 0.75 mm spacer in both sides to create two air gaps. Tighten the screws. Turn on the power and adjust the voltage to 50V ac. Measure I_1 , E_2 , I_2 and record the results in a table, similar to Table 2 below, in your logbook. Calculate the power delivered to lamp and record this in the table. The lamp power is given by the product of the RMS voltage and current of the lamp, $P_L = E_2 I_2$ Watts.

Turn off the power.

8. Repeat step 7 for the other values of the gaps and complete Table 2. Turn off the power by bringing down the voltage to zero.

Table 2

Gap (mm)	E_1 (V)	I_1 (A)	E_2 (Volt)	I_2 (A)	P_L (W)
0	50				
0.75	50				
1.5	50				
3	50				

PART II: Self and mutual inductance with coupling factors

2.1 Objectives

1. To measure the resistance and reactance of a coil
2. To calculate self-inductance of a coil
3. To observe how the reluctance of the magnetic circuit and number of the coils affects the inductance
4. To calculate the mutual inductance between two coils

2.2 Brief Theory

When a coil is connected to an AC voltage, the current is limited by the impedance Z of the coil. Z is comprised of resistance R of the coil and inductive reactance X_L .

$$Z = \frac{E}{I} = \sqrt{R^2 + X_L^2} \quad (4)$$

$$X_L = \sqrt{Z^2 - R^2} = 2\pi fL; \quad L = \frac{X_L}{2\pi f} \quad (5)$$

where L is the inductance of the coil. The inductance depends on the reluctance of the magnetic circuit. Fig 4 illustrates the method for finding L .

The self-inductance of a coil is also given by

$$L = \frac{N\phi}{I} \quad (6)$$

where flux ϕ in rms is calculated from equation (2).

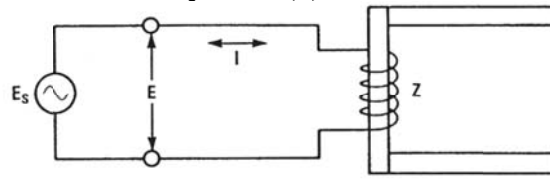


Fig. 4 Circuits for measuring the impedance Z

Additional equipment required

Wooden bar Additional resistance (220Ω) Fluke Multimeter AC Voltmeter

2.3 Procedure

1. Measure the DC resistances of coils A and B using the Fluke multimeter. Take one coil, making sure it is not connected to any power supply and no bar is inside the coil. Measure the resistance between terminals 3 and 4 ($N=2400$ turns) of the coil. Tabulate the values in a table like Table 3 in your logbook. Now measure the resistance between terminals 1 and 2 ($N=600$ turns) of the coil and note down the result in the same table. Repeat same for the second coil.

Table 3

Coil A : Resistance R (2400 turns)	
Coil A: Resistance R (600 turns)	
Coil B : Resistance R (2400 turns)	
Coil B : Resistance R (600turns)	

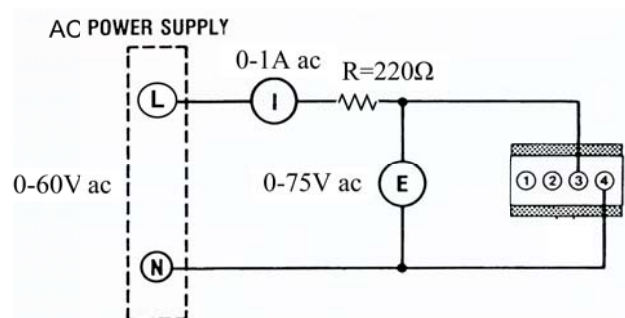


Fig 5 Circuit to determine Z , X_L and L of a coil with air core

2. Now connect the circuit to AC supply as shown in Fig. 5 with the 220 resistor connected in series with coil A. Turn on the power and adjust the current I to 0.20 A rms. Measure the voltage E across the coil and current I .

Turn off the power.

3. Use the same circuit arrangement as step 2, now with the wooden bar inserted inside the coil. Turn on the power and adjust the current I to 0.20 A ac. Measure the value of E .

4. Repeat step 3, now with a 133 mm laminated bar inside the coil. Turn on the power and adjust the current I to 0.20 A ac. Measure the value of E .
5. Tabulate the calculated inductances from measurements in 2, 3 and 4 using equations (4) and (5) in a table like **Table 4A** in your logbook. Include also the inductances calculated using equation 6 for each case.
6. Repeat steps 2, 3 and 4 for the 600 turns coil between terminals 1 & 2, as opposed to terminals 3 & 4, as used earlier. Tabulate the results in a table like **Table 4B** in your logbook.

Table 4A

Coil with 2400 turns	$Z (\Omega)$	$X_L (\Omega)$	$L (H)$	L from Eq 6
Air core				
Wooden bar inserted				
Laminated bar inserted				

Table 4B

Coil with 600 turns	$Z (\Omega)$	$X_L (\Omega)$	$L (H)$	L from Eq 6
Air core				
Wooden bar inserted				
Laminated bar inserted				

7. Calculate the ratio of inductances, $\frac{L_{2400 \text{ turns}}}{L_{600 \text{ turns}}}$ for each core.

Question 2 Explain why the coil inductances are different for the coil with different core materials.

Mutual inductance and the coefficient of coupling

The coupling of flux between two coupled coils should be high. The set-up of Fig. 6 is used to study this coupling between coils A and B.

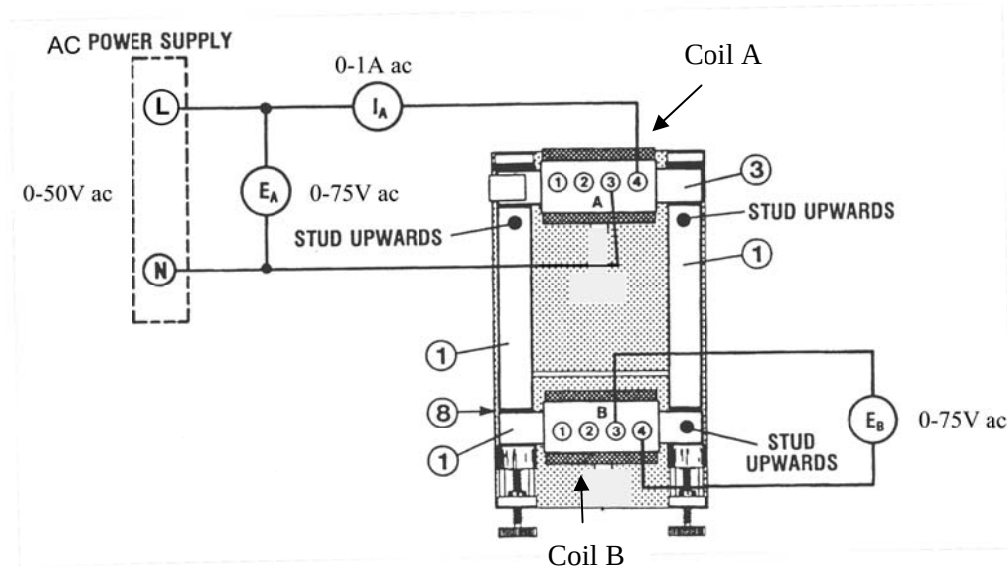


Fig. 6 circuit used to measure X_A and M_{AB} and k_{AB}

8. Set up the circuit of Fig. 6 using two 0.75 mm fibre spacers on the two sides close to coil B. Note that coil A and B are identical. Tighten the screw to minimize the remaining air gaps. Turn on the power and adjust voltage E_A to 50V ac. Measure I_A , E_B and calculate L_A , M_{AB} and K_{AB} using equation (5), (7)-(10). The self-inductance of coil A is found from equations 4 and 5. The mutual inductance between coils A and B is found from

$$X_{AB} = \frac{E_B}{I_A} \quad (7)$$

$$M_{AB} = \frac{X_{AB}}{2\pi f} \quad (8)$$

The flux linkage of coil $\phi_A = L_A I_A$. The flux linkage of coil B is found from equation 3, i.e.,

$$\phi_B = \frac{E_B}{4.44 f N_2} \quad (9)$$

The coupling coefficient between the two coils A and B is given by

$$k_{AB} = \frac{\phi_B}{\phi_A} \quad (10)$$

9. Reconnect the circuit of Figure 6 as shown in Figure 7, with the power supplied to coil B instead of coil A. Apply 50V to coil B. Measure I_B and E_A and calculate L_B , M_{AB} and K_{AB} . Use equations (5), (7)-(10) (with appropriate change of subscript) to calculate M_{AB} , ϕ_A and

k_{AB} . Note that double subscripts are interchangeable, i.e., $M_{AB} = M_{BA}$ and $k_{AB} = k_{BA}$, because the physical dimensions in Figures 6 and 7 are the same.

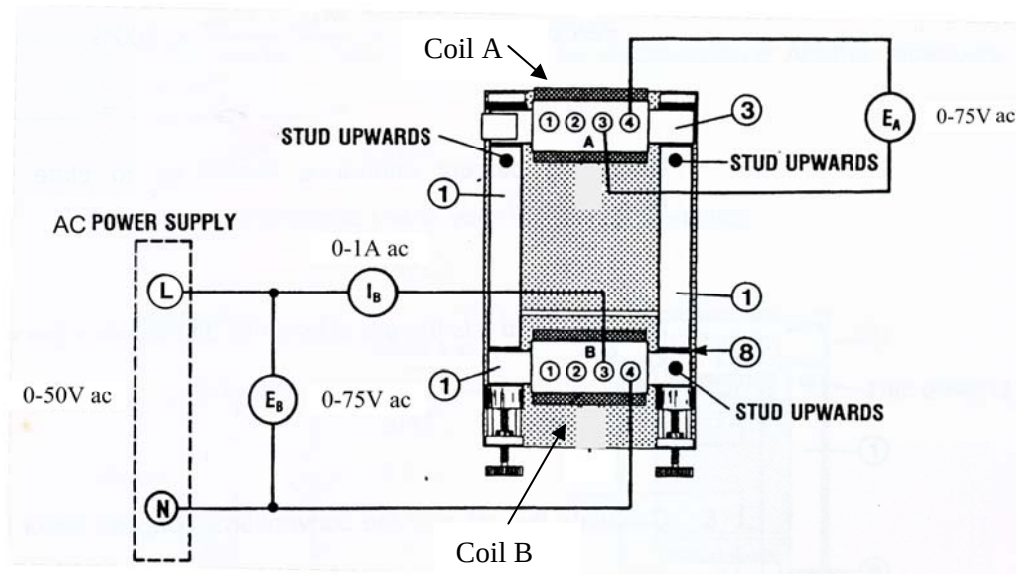


Fig. 7 Circuit used for measuring X_B and M_{BA} and k_{BA}

10. Calculate the coupling factor also from, $k_{AB} = \frac{M_{AB}}{\sqrt{L_A L_B}}$

Turn power off.

Disconnect the circuit and return all cables to their holders.